A PERFECT COMPLEX WHOSE COHOMOLOGY IS LOCALLY FREE ON A DENSE OPEN BECOMES OF TOR DIMENSION AT MOST ONE AFTER A SUITABLE ADMISSIBLE BLOWUP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0ESQ (sources/stacks-flat/pinned-original-source.tex line 13036).

Allowed standard prerequisites (editorial).

- The global Fitting ideals Fit_{i,k}(E) of a perfect complex and their independence of the representative. Same chapter, Lemma labelled lemma-fitting-ideals-complex, cited at proof lines 13067-13070. Identifies the ideal of minors with a globally defined ideal of O_X.
- Blowing up in a product of ideals and admissibility of the blowup in an invertible Fitting ideal. Divisors, Lemma labelled divisors-lemma-blowing-up-two-ideals, cited at proof lines 13079-13081, and Lemma labelled divisors-lemma-blowup-fitting-ideal, cited at proof lines 13081-13083 and 13092-13094. Shows that the product blowup dominates the individual admissible blowups and controls the tor dimension after the blowup.
- Domination of admissible blowups. Divisors, Lemma labelled divisors-lemma-dominate-admissible-blowups, cited at proof lines 13083-13085. Supplies the comparison with the proof that the dominating blowup is again U-admissible.

Source reference. Original lemma 0ESP: sources/stacks-flat/pinned-original-source.tex, source lines 12963–12963. Same-chapter pointer into the author environment lemma at source line 12963; invoked at proof line 13066 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080A: sources/stacks-chow/tags.txt, source lines 10175–10175. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13079 of the counted proof. It is

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Source reference. Original lemma 0ESN: sources/stacks-chow/tags.txt, source lines 18488–18488. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13080 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080N: sources/stacks-chow/tags.txt, source lines 10188–10188. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13083 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a scheme. Let $E \in D(\mathcal{O}_X)$ be perfect. Let $U \subset X$ be a scheme theoretically dense open subscheme such that $H^i(E|_U)$ is finite locally free of constant rank r_i for all $i \in \mathbf{Z}$. Then there exists a U-admissible blowup $b : X' \to X$ such that $H^i(Lb^*E)$ is a perfect $\mathcal{O}_{X'}$-module of tor dimension ≤ 1 for all $i \in \mathbf{Z}$.*

Proof. We will construct and study the blowup affine locally. Namely, suppose that $V \subset X$ is an affine open subscheme such that $E|_V$ can be represented by the complex

$$\mathcal{O}_V^{\oplus n_a} \xrightarrow{d_a} \ldots \xrightarrow{d_{b-1}} \mathcal{O}_V^{\oplus n_b}$$

Set $k_i = r_{i+1} - r_{i+2} + \ldots + (-1)^{b-i+1} r_b$. A computation which we omit show that over $U \cap V$ the rank of d_i is

$$\rho_i = -k_i + n_{i+1} - n_{i+2} + \ldots + (-1)^{b-i+1} n_b$$

in the sense that the cokernel of d_i is finite locally free of rank $n_{i+1} - \rho_i$. Let $\mathcal{I}_i \subset \mathcal{O}_V$ be the ideal generated by the minors of size $\rho_i \times \rho_i$ in the matrix of d_i.

On the one hand, comparing with Lemma 0ESP we see the ideal $\mathcal{I}_i$ corresponds to the global ideal $\mathrm{Fit}_{i,k_i}(E)$ which was shown to be independent of the choice of the complex representing $E|_V$. On the other hand, $\mathcal{I}_i$ is the $(n_{i+1} - \rho_i)$th Fitting ideal of $\mathrm{Coker}(d_i)$. Please keep this in mind.

We let $b : X' \to X$ be the blowing up in the product of the ideals $\mathrm{Fit}_{i,k_i}(E)$; this makes sense as locally on X almost all of these ideals are equal to the unit ideal (see above). This blowup dominates the blowups $b_i : X_i' \to X$ in the ideals $\mathrm{Fit}_{i,k_i}(E)$, see Divisors, Lemma 080A. By Divisors, Lemma 0ESN each b_i is a U-admissible blowup. It follows that b is a U-admissible blowup (tiny detail omitted; compare with the proof of Divisors, Lemma 080N). Finally, U is still a scheme theoretically dense open subscheme of X'. Thus after replacing X by X' we end up in the situation discussed in the next paragraph.

Assume $\mathrm{Fit}_{i,k_i}(E)$ is an invertible ideal for all i. Choose an affine open V and a complex of finite free modules representing $E|_V$ as above. It follows from Divisors, Lemma 0ESN that $\mathrm{Coker}(d_i)$ has tor dimension ≤ 1. Thus $\mathrm{Im}(d_i)$ is finite locally free as the kernel of a map from a finite locally free module to a finitely presented module of tor dimension ≤ 1. Hence $\mathrm{Ker}(d_i)$ is finite locally free as well (same argument). Thus the short exact sequence

$$0 \to \mathrm{Im}(d_{i-1}) \to \mathrm{Ker}(d_i) \to H^i(E)|_V \to 0$$

shows what we want and the proof is complete. □